Prashant Rahul

Home: prashantrahul.com
Email: prashantrahul141@protonmail.com
Github: github.com/prashantrahul141

Work experience

Cypherock, Gurugram *(Remote)* *Aug 2025 – Jan 2026*

Software Engineer Intern

- Implemented the Canton Network within STM32 embedded firmware for the X1 crypto hardware wallet, including writing real time encoders and decoders, hashing and signing Canton transactions.
- Ocassionally worked on the Electron side of things with typescript

Open source contributions

RTEMS - RTOS for Multiprocessor Systems [link]

- Enabled IMFS testing on BSPs with tight memory constraints, like the sparc/erc32 bsp
- Implemented POSIX compliant `ENAMETOOLONG` handling in `cpukit/libio` by validating path components against `NAME_MAX` during filesystem path evaluation

The Julia Programming Language [link]

- Improved performance in show() for UnitRange append! for AbstractVector
- Quality of life improvements when displaying empty ranges

Education

Bachelor's Degree in Computer Science *2027*

CV Raman Global University

Projects

evelin *(compiler, qbe, rust)* [link]

Evelin is a general purpose, statically typed, compiled language using the QBE backend with C FFI support.

zspie *(c, compiler, interpreter)* [link]

Zspie is a fast, easy, dynamic programming language, implemented in C with a virtual machine and opcode based execution.

cipi8 *(chip8, chip8-emulator, cpp)* [link]

C++ Chip-8 emulator with opcode, graphics, input, and timer support.

splax *(interpreter, lexer, parser)* [link]

A memory-safe, easy, dynamic programming language in Rust with a tree-walk interpreter executing directly on the AST.

dotfiles *(dotfiles, nix, nixos-configuration)* [link]

Nix configuration using flakes and home manager for all my machines.

Skills

Programming languages: Rust, C/C++, Python, Go, LLVM IR, QBE IR, Typescript, Javascript

Technologies & Tools: GNU/Linux, Nix/NixOS, Git, CMake, Make, Meson, Vi/Vim/Neovim, POSIX Shells, NodeJS

Concepts & Practices: Embedded Systems, RTOS, Device drivers, Firmware development, Operating systems, Cryptography Fundamentals, ARM ISA, Version Control, CI/CD